Miles Keiser

Berkeley, CA • miles.keiser@berkeley.edu • miles-miles2.github.io (Portfolio)

Education

University of California, Berkeley — Berkeley, CA
Electrical Engineering & Computer Sciences — Graduating 2029
GPA: 3.8

Del Lago Academy — Escondido, CA
Valedictorian Circle, National Merit Scholarship - Commended — 2025
Unweighted GPA: 4.0, Weighted GPA: 4.26 | SAT: 1540 (780 Math, 760 Verbal, 99+%)
AP Scores: Calculus BC: 5, Computer Science: 5, US Government: 5, English Lang: 4

Experience

SLICE Lab, Berkeley - Dr. Borivoje Nikolic — Berkeley. CA
Undergraduate Researcher — 08/2026 - Current
- Integrating a 3rd Party HyperRAM controller into Chipyard, Berkeley's open source SoC framework, and attaching it to TileLink Bus with Chisel.
- Verifying controller for use on future tapeouts
- Designing FPGA verification board in Altium to test HyperRAM implementation in hardware

Opto22 — Temecula, CA
FPGA Intern — Summer 2026
- Optimized access times between IMX6 processor and Cyclone IV FPGA
- Used Quartus toolchain and created bash tests scripts to qualify changes for production
- Reduced access time/bus cycle time: Theoretical Limit: 17 -> 9 cycles, Practical Limit: 17 -> 11 cycles

San Diego Supercomputing Center, UCSD — San Diego, CA
Computational Chemistry Intern — Summer 2024
- Tested and profiled DeePMD-kit (a software package that uses ML to speed up computational chemistry simulations) on different hardware tiers, including Expanse and Voyager supercomputers
- Created Docker containers of DeePMD-kit, allowing it to be run on the supercomputers

CureScience — San Diego, CA
CureScience Scholars Program Intern — Summer 2022, Summer 2023
- Predicted sensitivity of cancer cell lines using various machine learning techniques in Weka
- Created a combined mutation and gene pathway dataset with Python

Activities

Low Voltage Subteam, Formula Electric Berkeley — 09/2025 - Current
- Lead Engineer on Low Voltage Power Distribution Board (LVPDB) for 2027
    - LVPDB integrates overvoltage and overcurrent protection and distributes power to all other boards on the car. Schematic and layout in Altium.
    - Reduced board size vs. previous year by 51%
- Engineer on Autonomous Relay Control board for 2026
    - Handles Shutdown circuit in autonomous mode and powers Auto Emergency Brake

Combat Robotics Berkeley — 01/2026 - Current
- Designing chassis, armor, motor mount in CAD. Spec'd electronics package for a 3 lb combat robot (Fall 2026)
- Led CAD design of 3D printed 1 lb combat robot in Onshape (Spring 2026)

AI Mobile app to help the visually impaired identify currency — Summer 2023
- With help from mentor Amit Gupta, PhD, built an iOS app using Swift and Xcode with data crowdsourcing functionality

Skills

HDL: Chisel, Verilog | **Tools:** Quartus - Waveform debugging, Altium, KiCAD | **Languages:** Python, C, Bash, Java
Onshape | Blender